# Beyond Entropy: Region Confidence Proxy for Wild Test-Time Adaptation

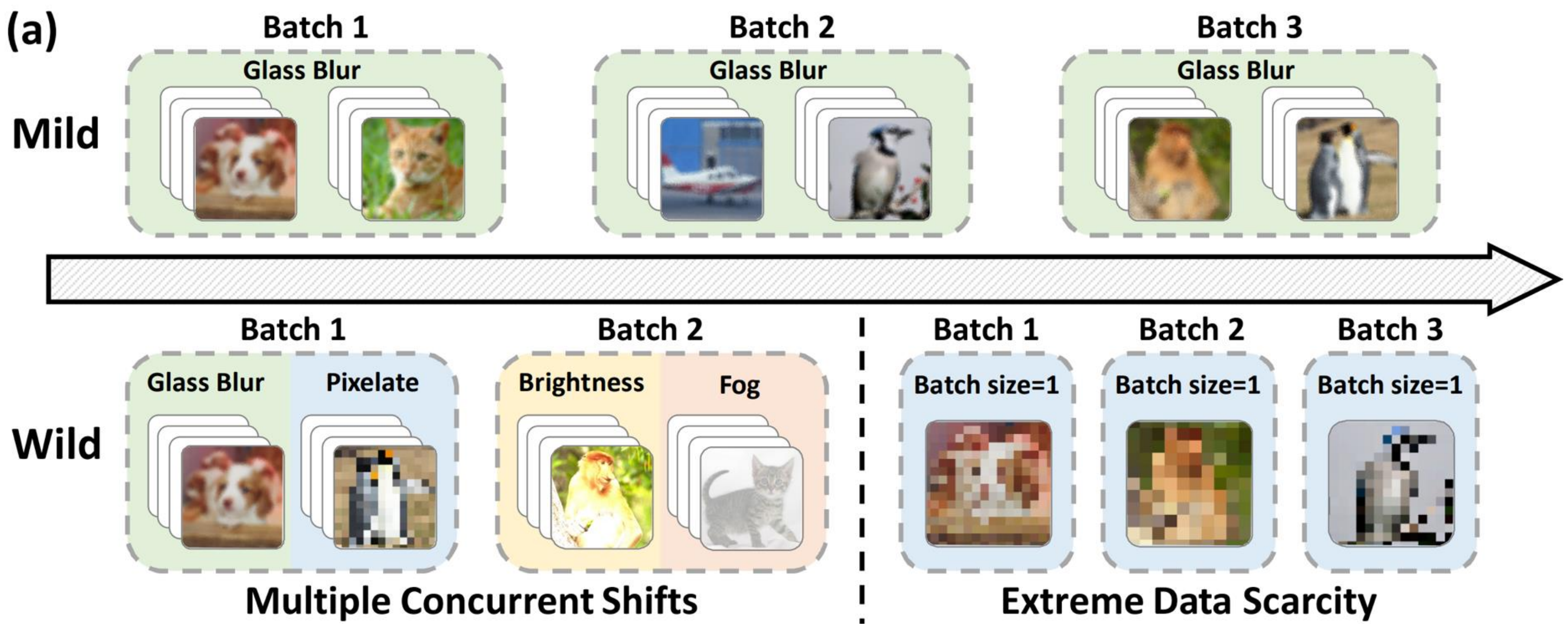
ICML 2025

Hyunoh Kang

HAI
Human-centered AI

# Introduction

- TTA methods have achieved promising results under mild conditions
- But, they show significant performance drops in wild scenarios
  - ➢ focus on developing selection criteria to leverage reliable samples




HAI
Human-centered AI

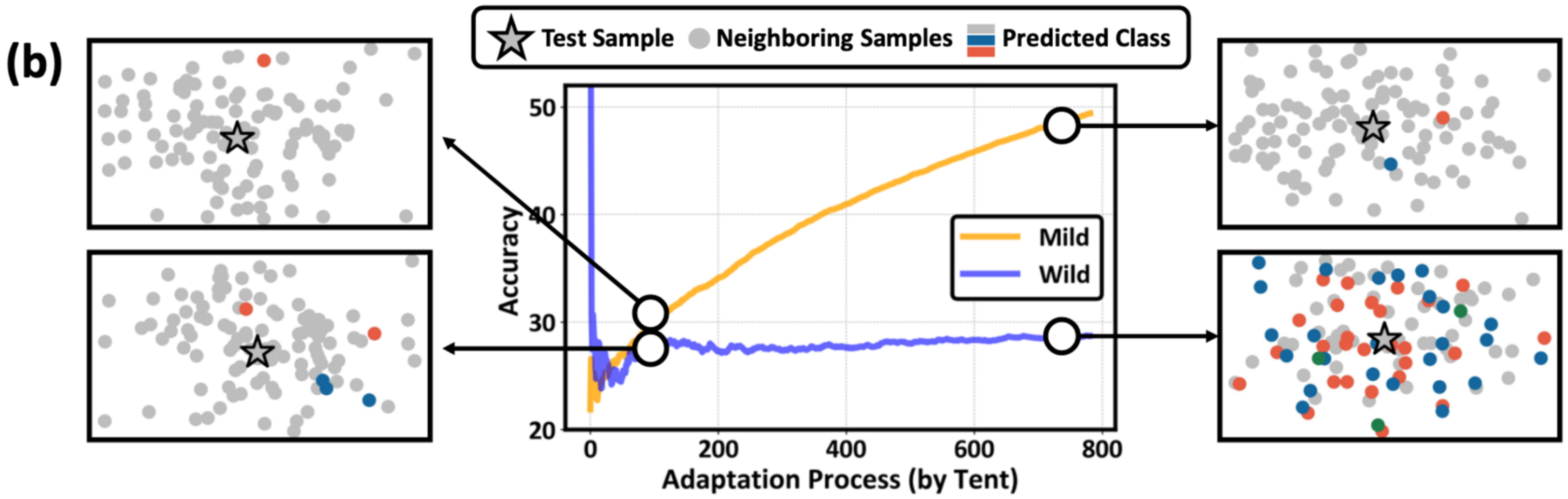
# Introduction

- TTA methods have achieved promising results under mild conditions
- But, they show significant performance drops in wild scenarios
  - focus on developing selection criteria to leverage reliable samples
    - SAR (Niu et al., 2023) excluded samples with high entropy
    - DeYO (Lee et al., 2024) filtered out samples with sensitive prediction changes under image transformation

  - This paper aims to approach the more fundamental problem of entropy minimization
  - Not what to choose(criteria), but **what to minimize(objective)**

# Introduction

- In wild scenes, we observe a notable instability in feature space
- Feature space should be grouped with samples that have semantically similar meanings
  - Enhance Region confidence



# Preliminaries for Wild Test-Time Adaptation

- Practical Scenario in Wild TTA
  - Limited data stream: batch size = 1
  - Mixed testing domain: target domain consists of k different sub-domains
  - Imbalanced label shift: the test label distribution is imbalanced and shifts over time according to a function $Q_t(y)$

  - Existing solutions: Entropy Minimization + selection criteria

# Preliminaries for Wild Test-Time Adaptation

- Practical Scenario in Wild TTA
  - Limited data stream: batch size = 1
  - Mixed testing domain: target domain consists of k different sub-domains
  - Imbalanced label shift: the test label distribution is imbalanced and shifts over time according to a function $Q_t(y)$

  - Existing solutions: Entropy Minimization + selection criteria

# Exploring Entropy Minimization via Local Consistency

- EM's effectiveness heavily relies on the <u>local consistency</u> to extend sample-wise confidence to a regional scale
  - * Local consistency: nearby points share the similar prediction

- local consistency
  - O: optimization for individual points align with the overall direction, magnifying the local effects
  - X: introducing noise that hinders adaptation in the affected region

➢ it is crucial to evaluate the reliability of entropy minimization in **preserving local consistency** under wild scenes.

HAI
Human-centered AI

# Exploring Entropy Minimization via Local Consistency

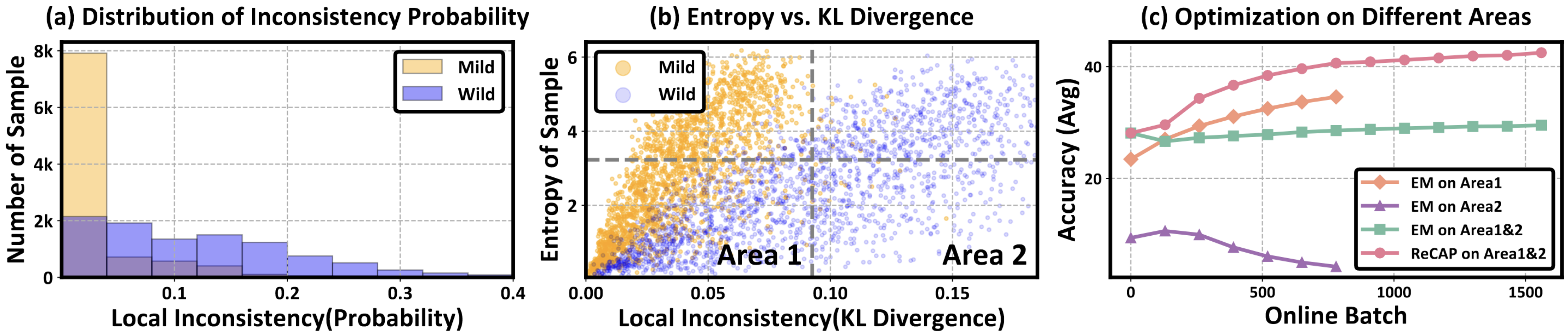



*Figure 2.* Local consistency during the entropy minimization process under mild and wild (imbalanced label shift) scenarios. Consistency is measured by prediction discrepancies between each sample and its 256 neighboring samples. (a) shows the probability of inconsistent predictions in neighbors. (b) records the entropy and average KL Divergence between prediction probabilities of samples and their neighbors. (c) investigates adaptation performance when using samples with varying levels of local consistency. All experiments are conducted on ImageNet-C of Gaussian noise with ResNet50. 'EM' denotes entropy minimization and 'ReCAP' denotes our method.

HAI
Human-centered AI

# From Sample Confidence to Region Confidence

- Introduce a novel objective called **region confidence** to replace vanilla entropy
  - optimizes both region-wise confidence and stability simultaneously
  - improving global optimization efficiency
- Mathematical definition:

$$\mathcal{L}_{RC}(x) = \underbrace{\int_{\Omega} \mathcal{L}_{\text{ent}}(\tilde{x}) d\tilde{x}}_{\text{Bias term}} + \lambda \underbrace{\int_{\Omega} \mathcal{D}_{KL}\left(p_\theta(x) \| p_\theta(\tilde{x})\right) d\tilde{x}}_{\text{Variance term}},$$

HAI
Human-centered AI

# Region Confidence Proxy

- There are still two considerable challenges to optimize region confidence
  1. Uncertain region scope
     - The static regions fix the sample location, making it difficult to determine an appropriate local scope
  2. Heavy computational burden
     - Previous term is intractable, relying on extensive sampling and additional model forwards for measurement
- Propose **Region Confidence Adaptive Proxy (ReCAP)**
  - Probabilistic region modeling mechanism
  - Efficient proxy for optimizing region confidence

# Region Confidence Proxy

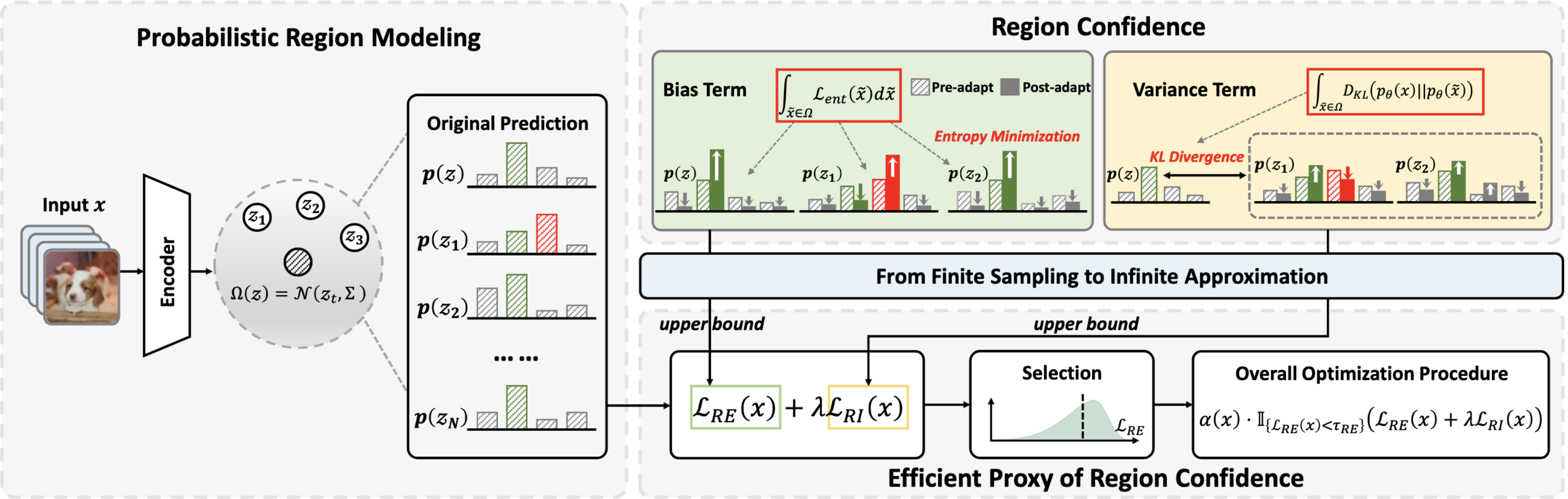



*Figure 3.* Overview of our ReCAP. ReCAP performs probabilistic modeling to determine local regions in the latent space (Sec. 4.1). We further derives two closed-form upper bounds for the intractable bias and variance terms via a finite-to-infinite asymptotic approximation, offering an efficient proxy for optimizing Region Confidence without the lengthy sampling process (Sec. 4.2 & 4.3).

HAI
Human-centered AI

# Probabilistic Region Modeling

- Different directions in feature space can imply potentials of valuable semantic changes
  - Model local regions **in feature space** to flexibly capture rich semantic information
    - Select the latent space extracted by the backbone
    - Avoid class mixing

- Mathematical Definition

$$\mathcal{L}_{ent}(x) = -\boxed{p_\theta(x)} \cdot \log \boxed{p_\theta(x)} \rightarrow p_\theta(z)_i = (\operatorname{softmax}(Az+b))_i = \frac{e^{a_i \cdot z + b_i}}{\sum_{j=1}^{C} e^{a_j \cdot z + b_j}}$$

  - i: class
  - $A \in \mathbb{R}^{C \times d}$ $b \in \mathbb{R}^{C}$: the linear and bias coefficients of the affine layer in the classifier

HAI
Human-centered AI

# Probabilistic Region Modeling

- Local region as a static range -> a probabilistic representation following a **multivariate Gaussian distribution**

$$\Omega\left(z_t\right) := \mathcal{N}\left(z_t, \tau \cdot \Sigma\right), \Sigma = \mathrm{Diag}\left(\mathrm{Var}_{\mathcal{D}_{\mathcal{S}}}(z)\right)$$

- $\Omega(z_t)$: the local region of the $t$-th test batch $z_t$
- $\Sigma$: the diagonal matrix of variance on a small set of source data (e.g.500 is enough)
- $\tau$: a hyper-parameter to control the scope.

# Efficient Metric of Region Confidence

- the computation of two terms
  - depends on an infinite number of potential samples from the distribution
  - requires extensive sampling to capture the statistical information

  ➢ Computational overhead and memory usage increase almost linearly with the number of sampling

➢ Approximation

# Efficient Metric of Region Confidence

**Lemma 4.1.** ***(Bias Term under Finite Sampling)*** *Given N features $z_1, \ldots, z_N$ drawn from the local region and their corresponding probabilities: $p_\theta(z_1), \ldots, p_\theta(z_N)$. The bias term on these features satisfies the following inequality:*

$$\sum_{i=1}^{N} \mathcal{L}_{ent}(p_\theta(z_i)) \leqslant -\sum_{i=1}^{C} \frac{\sum_{k=1}^{N} p_\theta(z_k)_i}{N} \cdot \left( \sum_{j=1}^{N} \log p_\theta\left(z_j\right)_i \right), \tag{5}$$

HAI
Human-centered AI

# Efficient Metric of Region Confidence

**Lemma 4.2.** *(Upper Bound of Negative Log-Likelihood) Given a feature $z$ and its local region $\Omega$ which follows a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$. The expected value of the logarithm of the predicted probability for the i-th class satisfies the following inequality:*

$$
\begin{aligned}
&- \mathbb{E}_{z\sim\mathcal{N}(\mu,\Sigma)} \log p_\theta(z)_i \\
&\leq \log \sum_{j=1}^{C} e^{(a_j - a_i)\cdot\mu + (b_j - b_i) + \frac{1}{2}(a_j - a_i)\Sigma(a_j - a_i)^\top},
\end{aligned} \tag{6}
$$

HAI
Human-centered AI

# Efficient Metric of Region Confidence

- $L_{RE}$: Regional Entropy

**Proposition 4.3.** *(Efficient Metric of Bias Term) Given a feature $z$ and its local region $\Omega$ which follows a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$. The expectation of entropy loss over the entire distribution has a closed-form upper bound:*

$$\begin{aligned}\mathbb{E}_\Omega[\mathcal{L}_{ent}] &= -\,\mathbb{E}_{\tilde{z}\sim\mathcal{N}(z,\Sigma)}\sum_{i=1}^{C} p_\theta(\tilde{z})_i \log p_\theta(\tilde{z})_i \\ &\leqslant \sum_{j=1}^{C} \frac{e^{a_j\cdot z+b_j+\frac{1}{2}a_j\Sigma a_j^\top}}{\sum_{k=1}^{C} e^{a_k\cdot z+b_k+\frac{1}{2}a_k\Sigma a_k^\top}} \log\sum_{i=1}^{C} e^{(b_i-b_j)} \\ &\quad \cdot e^{(a_i-a_j)\cdot z+\frac{1}{2}(a_i-a_j)\Sigma(a_i-a_j)^\top} \triangleq \mathcal{L}_{RE}\,.\end{aligned} \tag{7}$$

# Efficient Metric of Region Confidence

- $L_{RI}$: Regional Instability

**Proposition 4.4.** ***(Efficient Metric of Variance Term)*** *Given a feature $z$ and its local region $\Omega$ which follows a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$. The expectation of Kullback-Leibler divergence between the output probability over this distribution and the probability at its center has a upper bound:*

$$
\begin{aligned}
&E_{\tilde{z}\sim\mathcal{N}(z,\Sigma)} KL\left(p_\theta(z) \| p_\theta(\tilde{z})\right) \\
&\leq \sum_{j=1}^{C} \frac{e^{a_j \cdot z + b_j}}{\sum_{k=1}^{C} e^{a_k \cdot z + b_k}} \cdot \log \sum_{i=1}^{C} \frac{e^{a_i \cdot z + b_i}}{\sum_{k=1}^{C} e^{a_k \cdot z + b_k}} \qquad (8) \\
&\cdot e^{\frac{1}{2}(a_i - a_j)\Sigma(a_i - a_j)^\top} \triangleq \mathcal{L}_{RI},
\end{aligned}
$$

HAI
Human-centered AI

# Overall Procedure of ReCAP

$$\min_{\theta} \alpha(x) \cdot \mathbb{I}_{\{\mathcal{L}_{RE}(x)<\tau_{RE}\}}(\mathcal{L}_{RE}(x) + \lambda\mathcal{L}_{RI}(x)),$$
$$\text{where } \alpha(x) \triangleq \frac{1}{exp(\mathcal{L}_{RE}(x) - \mathcal{L}_0)}, \tag{9}$$

HAI
Human-centered AI

# Overall Procedure of ReCAP

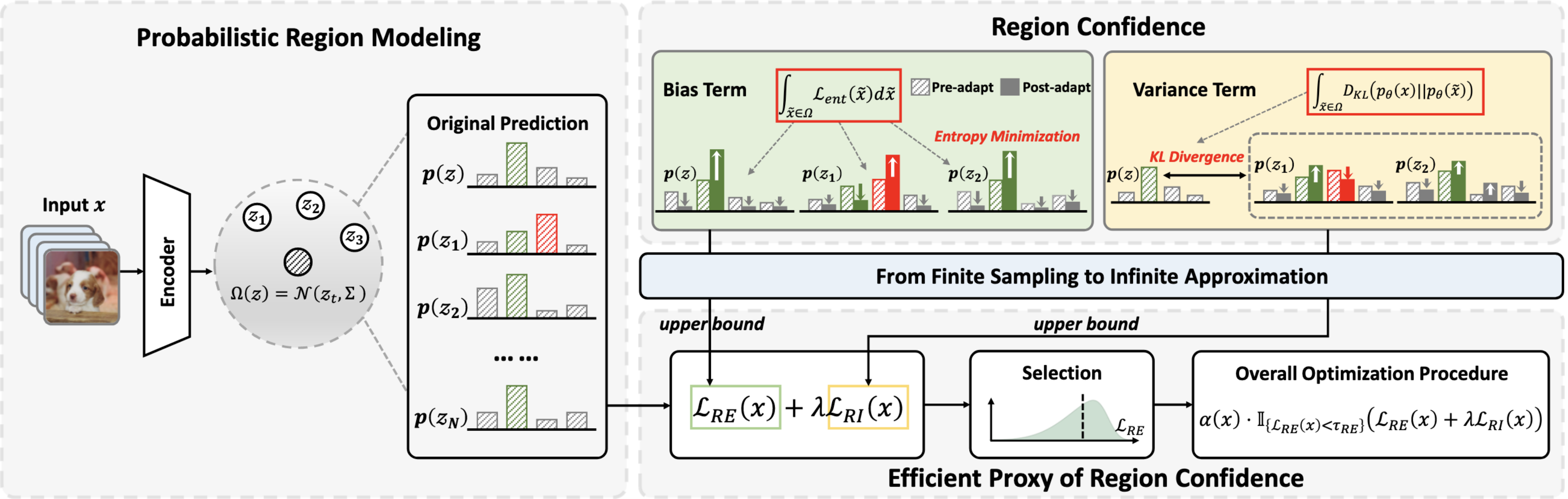



*Figure 3.* Overview of our ReCAP. ReCAP performs probabilistic modeling to determine local regions in the latent space (Sec. 4.1). We further derives two closed-form upper bounds for the intractable bias and variance terms via a finite-to-infinite asymptotic approximation, offering an efficient proxy for optimizing Region Confidence without the lengthy sampling process (Sec. 4.2 & 4.3).

HAI
Human-centered AI

# Ablation studies

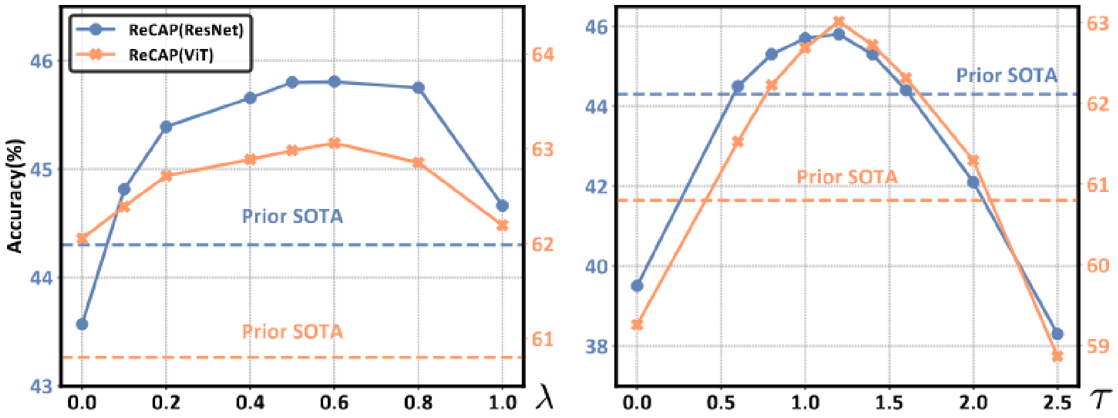



*Figure 4.* (a) Performance with varying strengths $\lambda$ of the variance term. (b) Performance with different ranges $\tau$ of the local region.

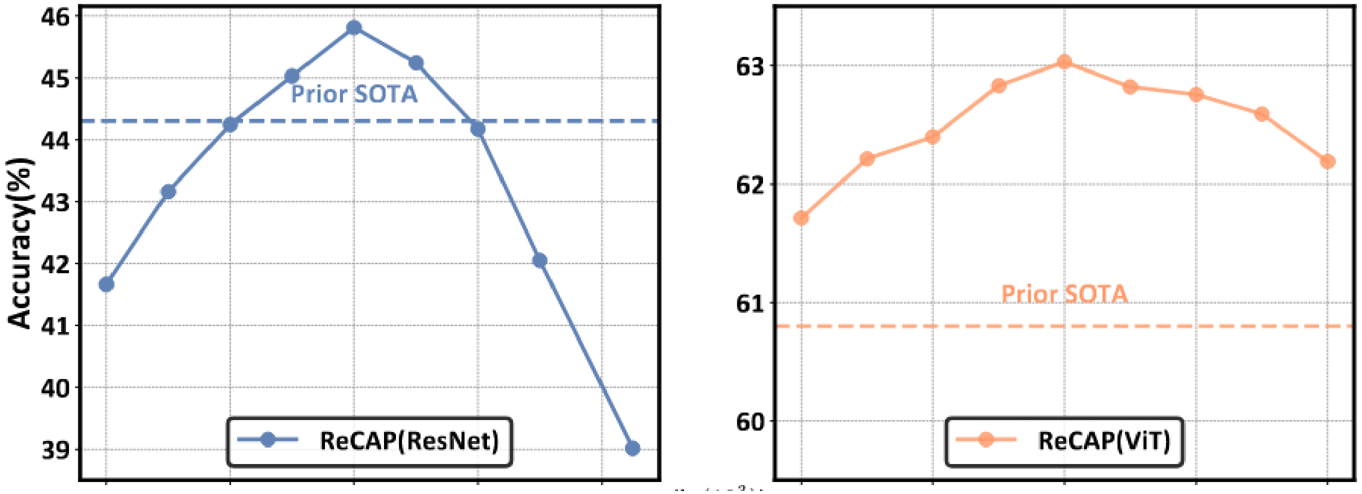



*Figure 7.* Performance under different selection boundary $\tau_{RE}$ for ResNet and ViT on ImageNet-C under label shifts.

# Ablation studies

*Table 7.* Effects of components in ReCAP. For a fair comparison, '+Vanilla Entropy' uses entropy-based selection and weighting.

| Component | | | Corruption Category | | | | Average |
|---|---|---|---|---|---|---|---|
| Vanilla Entropy | $\mathcal{L}_{RE}$ in Eq. 7 | $\mathcal{L}_{RI}$ in Eq. 8 | Noise | Blur | Weather | Digital | |
| ✔ | | | 24.9 | 19.6 | 41.5 | 37.8 | 31.4 |
| | ✔ | | 41.9 | 29.9 | 56.4 | 44.5 | 43.3 |
| ✔ | | ✔ | 41.3 | 26.8 | 55.0 | 47.3 | 42.7 |
| | ✔ | ✔ | **42.9** | **31.0** | **57.1** | **51.2** | **45.8** |

HAI
Human-centered AI

# Experiments

*Table 1.* Comparisons with state-of-the-art methods on ImageNet-C (severity level 5) under **Limited Batch Size = 1** regarding Accuracy (%). We report mean performance over 3 independent runs. The best results are in bold and the second-best are in underline.

| Model+Method | Noise | | | Blur | | | | Weather | | | | Digital | | | | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Gauss. | Shot | Impul. | Defoc. | Glass | Motion | Zoom | Snow | Frost | Fog | Brit. | Contr. | Elastic | Pixel | JPEG | |
| ResNet50 | 18.0 | 19.8 | 17.9 | 19.8 | 11.4 | 21.4 | 24.9 | 40.4 | 47.3 | 33.6 | 69.3 | 36.3 | 18.6 | 28.4 | 52.3 | 30.6 |
| • MEMO | 18.5 | 20.5 | 18.4 | 17.1 | 12.6 | 21.8 | 26.9 | 40.4 | 47.0 | 34.4 | 69.5 | 36.5 | 19.2 | 32.1 | 53.3 | 31.2 |
| • DDA | 42.4 | 43.3 | 42.3 | 16.6 | 19.6 | 21.9 | 26.0 | 35.7 | 40.1 | 13.7 | 61.2 | 25.2 | 37.5 | 46.6 | 54.1 | 35.1 |
| • Tent | 2.5 | 2.9 | 2.5 | 13.5 | 3.6 | 18.6 | 17.6 | 15.3 | 23.0 | 1.4 | 70.4 | 42.2 | 6.2 | 49.2 | 53.8 | 21.5 |
| • EATA | 24.9 | 28.0 | 25.8 | 18.3 | 17.0 | 31.2 | 29.8 | 42.5 | 44.1 | 41.3 | 70.9 | 44.2 | 27.6 | 46.8 | 55.4 | 36.5 |
| • SAR | 25.5 | 28.0 | 24.9 | 18.7 | 16.3 | 28.6 | 31.4 | 46.2 | 44.9 | 33.4 | 72.8 | 44.3 | 15.3 | 47.1 | 56.1 | 35.6 |
| • DeYO | 41.2 | 44.3 | 42.5 | 22.4 | 24.7 | 41.8 | 21.9 | 54.8 | 51.6 | 21.9 | 73.1 | 53.2 | 48.5 | 59.8 | 59.6 | 44.1 |
| • ReCAP (Ours) | **42.5** | 44.4 | **42.9** | 19.4 | 25.0 | 42.2 | 44.0 | 49.7 | 52.4 | **57.5** | 72.9 | 53.6 | 29.5 | 60.4 | 60.0 | 46.4 |
| • ReCAP+SAR | 41.7 | 44.5 | 40.6 | 24.8 | 25.8 | **44.0** | **47.0** | **56.2** | **53.0** | 52.8 | 73.4 | 54.6 | 48.8 | **61.7** | **60.7** | 48.6 |
| • ReCAP+DeYO | **42.5** | **44.8** | 42.8 | **25.9** | **27.2** | 43.7 | 44.9 | 55.8 | 52.8 | 51.9 | **73.5** | **54.8** | **50.9** | 61.5 | **60.7** | **48.9** |
| Vitbase | 9.5 | 6.7 | 8.2 | 29.0 | 23.4 | 33.9 | 27.1 | 15.9 | 26.5 | 47.2 | 54.7 | 44.1 | 30.5 | 44.5 | 47.8 | 29.9 |
| • MEMO | 21.6 | 17.3 | 20.6 | 37.1 | 29.6 | 40.4 | 34.4 | 24.9 | 34.7 | 55.1 | 64.8 | 54.9 | 37.4 | 55.4 | 57.6 | 39.1 |
| • DDA | 41.3 | 41.1 | 40.7 | 24.4 | 27.2 | 30.6 | 26.9 | 18.3 | 27.5 | 34.6 | 50.1 | 32.4 | 42.3 | 52.2 | 52.6 | 36.1 |
| • Tent | 42.2 | 1.0 | 43.3 | 52.4 | 48.2 | 55.5 | 50.5 | 16.5 | 16.9 | 66.4 | 74.9 | 64.7 | 51.6 | 67.0 | 64.3 | 47.7 |
| • EATA | 30.1 | 24.6 | 34.2 | 44.3 | 39.6 | 48.4 | 42.4 | 38.1 | 46.0 | 60.7 | 65.8 | 61.2 | 46.7 | 57.8 | 59.5 | 46.6 |
| • SAR | 42.7 | 39.5 | 41.9 | 54.6 | 51.2 | 58.3 | 54.4 | 60.2 | 54.7 | 70.3 | 75.9 | 66.8 | 58.4 | 69.5 | 66.3 | 57.6 |
| • DeYO | 53.4 | 50.4 | 55.0 | 58.7 | 59.5 | 64.5 | 52.5 | 68.1 | 66.3 | 73.8 | 78.3 | 67.9 | 68.9 | 73.8 | 70.8 | 64.1 |
| • ReCAP (Ours) | 53.5 | **56.7** | **56.9** | 59.2 | 60.5 | 65.3 | 64.0 | 69.6 | 67.2 | 74.1 | 78.4 | 64.6 | 70.2 | 74.4 | 71.5 | 65.7 |
| • ReCAP+SAR | **55.3** | 56.2 | 56.5 | **60.0** | **61.2** | **66.5** | **65.2** | **69.8** | **68.0** | **74.5** | **78.6** | **68.4** | **71.0** | **74.9** | **71.6** | **66.5** |
| • ReCAP+DeYO | 54.4 | 55.3 | 55.5 | 59.8 | 61.1 | 65.2 | 64.9 | 69.3 | 67.9 | 73.9 | 78.4 | 67.1 | 70.7 | 74.4 | 71.0 | 65.9 |

*Table 2.* Comparisons with SOTA methods on ImageNet-C (severity level 5, 4) under **Mixed Testing Domain**.

| Model+Method | Level=5 | Level=4 | Average |
|---|---|---|---|
| ResNet50 | 30.6 | 42.7 | 36.7 |
| • MEMO | 31.2 | 43.0 | 37.1 |
| • DDA | 35.1 | 43.6 | 39.4 |
| • Tent | 13.4 | 20.6 | 17.0 |
| • EATA | 38.1 | 47.7 | 42.9 |
| • SAR | 38.3 | 48.6 | 43.5 |
| • DeYO | 38.6 | 50.2 | 44.4 |
| • ReCAP (Ours) | 41.5 | 51.2 | 46.4 |
| • ReCAP+SAR | 42.1 | 51.9 | 47.0 |
| • ReCAP+DeYO | **42.2** | **52.4** | **47.3** |
| VitBase | 29.9 | 42.9 | 36.4 |
| • MEMO | 39.1 | 51.3 | 45.2 |
| • DDA | 36.1 | 45.1 | 40.6 |
| • Tent | 16.5 | 64.3 | 40.4 |
| • EATA | 55.7 | 63.7 | 59.7 |
| • SAR | 57.1 | 64.9 | 61.0 |
| • DeYO | 59.4 | 66.8 | 63.1 |
| • ReCAP (Ours) | 59.4 | 67.1 | 63.3 |
| • ReCAP+SAR | **60.0** | **67.2** | **63.6** |
| • ReCAP+DeYO | 59.8 | 67.0 | 63.4 |

# Experiments

*Table 3.* Comparisons with state-of-the-art methods on ImageNet-C (severity level 5) under **Imbalanced Label Shift**.

| Model+Method | Noise | | | Blur | | | | Weather | | | | Digital | | | | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Gauss. | Shot | Impul. | Defoc. | Glass | Motion | Zoom | Snow | Frost | Fog | Brit. | Contr. | Elastic | Pixel | JPEG | |
| ResNet50 | 17.9 | 19.9 | 17.9 | 19.7 | 11.3 | 21.3 | 24.9 | 40.4 | 47.4 | 33.6 | 69.2 | 36.3 | 18.7 | 28.4 | 52.2 | 30.6 |
| • MEMO | 18.4 | 20.6 | 18.4 | 17.1 | 12.7 | 21.8 | 26.9 | 40.7 | 46.9 | 34.8 | 69.6 | 36.4 | 19.2 | 32.2 | 53.4 | 31.3 |
| • DDA | **42.5** | 43.4 | 42.3 | 16.5 | 19.4 | 21.9 | 26.1 | 35.8 | 40.2 | 13.7 | 61.3 | 25.2 | 37.3 | 46.9 | 54.3 | 35.1 |
| • Tent | 2.6 | 3.3 | 2.7 | 13.9 | 7.9 | 19.5 | 28.7 | 16.5 | 21.9 | 1.8 | 70.5 | 42.2 | 6.6 | 49.4 | 53.7 | 22.8 |
| • EATA | 27.2 | 28.5 | 28.4 | 15.1 | 16.7 | 24.6 | 25.5 | 32.5 | 32.2 | 40.0 | 66.5 | 33.2 | 24.1 | 42.2 | 38.6 | 31.7 |
| • SAR | 34.0 | 36.7 | 36.2 | 21.8 | 20.9 | 33.2 | 32.4 | 38.7 | 45.6 | 50.6 | 72.9 | 46.8 | 14.3 | 52.2 | 56.8 | 39.5 |
| • DeYO | 41.7 | 44.0 | 42.5 | 23.4 | 23.9 | 41.3 | 13.0 | 53.9 | 52.2 | 38.6 | 73.1 | 52.3 | 46.8 | 59.3 | 59.1 | 44.3 |
| • ReCAP (Ours) | 42.0 | 44.1 | 42.7 | 19.8 | 24.3 | 39.7 | 40.2 | 46.0 | 52.2 | 57.3 | 73.1 | 52.4 | 33.7 | 59.4 | 59.5 | 45.8 |
| • ReCAP+SAR | 42.2 | 44.4 | 42.7 | 24.1 | 24.3 | 41.6 | **43.8** | 51.6 | 52.4 | 56.8 | 73.1 | 52.9 | 38.9 | 60.2 | 59.4 | 47.2 |
| • ReCAP+DeYO | **42.5** | **44.5** | **43.3** | **25.8** | **26.7** | **43.3** | 39.1 | **54.2** | **53.2** | **59.3** | **73.4** | **53.8** | **49.2** | **61.4** | **60.3** | **48.7** |
| Vitbase | 9.4 | 6.7 | 8.3 | 29.1 | 23.4 | 34.0 | 27.0 | 15.8 | 26.3 | 47.4 | 54.7 | 43.9 | 30.5 | 44.5 | 47.6 | 29.9 |
| • MEMO | 21.6 | 17.4 | 20.6 | 37.1 | 29.6 | 40.6 | 34.4 | 25.0 | 34.8 | 55.2 | 65.0 | 54.9 | 37.4 | 55.5 | 57.7 | 39.1 |
| • DDA | 41.3 | 41.3 | 40.6 | 24.6 | 27.4 | 30.7 | 26.9 | 18.2 | 27.7 | 34.8 | 50.0 | 32.3 | 42.2 | 52.5 | 52.7 | 36.2 |
| • Tent | 32.7 | 1.4 | 34.6 | 54.4 | 52.3 | 58.2 | 52.2 | 7.7 | 12.0 | 69.3 | 76.1 | 66.1 | 56.7 | 69.4 | 66.4 | 47.3 |
| • EATA | 35.8 | 34.8 | 36.8 | 45.1 | 47.3 | 49.3 | 47.8 | 56.6 | 55.5 | 62.1 | 72.3 | 21.6 | 56.0 | 64.6 | 63.7 | 50.0 |
| • SAR | 48.2 | 48.7 | 49.0 | 55.4 | 54.5 | 59.2 | 54.3 | 55.8 | 54.5 | 70.0 | 76.9 | 66.1 | 62.2 | 70.2 | 66.5 | 59.4 |
| • DeYO | 53.0 | 34.4 | 48.8 | 57.6 | 58.5 | 63.3 | 35.4 | 67.4 | 66.0 | 73.0 | **77.7** | 66.6 | 68.1 | **73.1** | 69.8 | 60.8 |
| • ReCAP (Ours) | 53.1 | 38.5 | 49.6 | 57.3 | **59.0** | **63.8** | 60.7 | **67.8** | 66.3 | 72.9 | **77.7** | 66.8 | 68.2 | 73.0 | **70.0** | 63.0 |
| • ReCAP+SAR | **53.2** | 42.7 | 50.9 | **58.0** | 58.9 | **63.8** | **60.9** | 67.6 | 66.1 | **73.2** | 77.6 | **67.6** | **68.3** | **73.1** | 69.8 | 63.4 |
| • ReCAP+DeYO | 53.1 | **53.9** | **54.1** | 57.4 | 58.8 | 63.6 | 60.6 | **67.8** | **66.4** | 72.9 | **77.7** | 67.1 | **68.3** | **73.1** | 69.8 | **64.3** |

HAI
Human-centered AI

# Experiments

| Model+Method | Limited Batch Size = 1 | | | Imbalanced Label Shift | | | Average |
|---|---|---|---|---|---|---|---|
| | ResNet | VitBase | Avg | ResNet | VitBase | Avg | |
| No-Adapt Model | 40.8 | 43.1 | 42.0 | 40.8 | 43.1 | 42.0 | 42.0 |
| • Tent (Wang et al., 2020) | 43.2 | 43.8 | 43.5 | 42.4 | 46.8 | 44.6 | 44.1 |
| • EATA (Niu et al., 2022) | 44.1 | 52.5 | 48.3 | 42.1 | 50.5 | 46.3 | 47.3 |
| • SAR (Niu et al., 2023) | 46.7 | 55.5 | 51.1 | 44.3 | 54.4 | 49.4 | 50.2 |
| • DeYO (Lee et al., 2024) | 48.1 | 59.2 | 53.7 | 46.7 | 58.5 | 52.6 | 53.1 |
| • ReCAP (Ours) | **51.5**$_{\pm 0.5}$ | **61.1**$_{\pm 0.6}$ | **56.3**$_{\pm 0.5}$ | **49.6**$_{\pm 0.2}$ | **60.4**$_{\pm 0.2}$ | **55.0**$_{\pm 0.2}$ | **55.7**$_{\pm 0.4}$ |

*Table 5.* Comparisons with state-of-the-art methods on **ImageNet-R** under **Limited Batch Size = 1** & **Imbalanced Label Shift** regarding Accuracy (%). We report mean&std over 3 independent runs. The best results are in bold.

| Model+Method | Limited Batch Size = 1 | | | Imbalanced Label Shift | | | Average |
|---|---|---|---|---|---|---|---|
| | ResNet | VitBase | Avg | ResNet | VitBase | Avg | |
| No-Adapt Model | 43.5 | 44.3 | 43.9 | 43.5 | 44.3 | 43.9 | 43.9 |
| • Tent (Wang et al., 2020) | 43.9 | 50.6 | 47.3 | 43.7 | 50.1 | 46.9 | 47.1 |
| • EATA (Niu et al., 2022) | 44.2 | 49.5 | 46.9 | 43.5 | 51.6 | 47.6 | 47.2 |
| • SAR (Niu et al., 2023) | 45.2 | 52.8 | 49.0 | 44.7 | 53.9 | 49.3 | 49.2 |
| • DeYO (Lee et al., 2024) | 45.8 | 58.5 | 52.2 | 45.2 | 57.1 | 51.2 | 51.7 |
| • ReCAP (Ours) | **48.0**$_{\pm 0.2}$ | **59.2**$_{\pm 0.9}$ | **53.6**$_{\pm 0.6}$ | **47.7**$_{\pm 0.2}$ | **58.5**$_{\pm 0.6}$ | **53.1**$_{\pm 0.4}$ | **53.4**$_{\pm 0.5}$ |

*Table 6.* Comparisons with state-of-the-art methods on **VisDA-2021** under **Limited Batch Size = 1** & **Imbalanced Label Shift** regarding Accuracy (%). We report mean&std over 3 independent runs. The best results are in bold.

# Experiments

*Table 4.* Efficiency comparison of various methods. We assess TTA approaches for processing 50,000 images in Gaussian corruption type, using a single Nvidia RTX 4090 GPU.

| Method | Forward | Backward | Other computation | Time |
|---|---|---|---|---|
| No-adapt | 50,000 | N/A | N/A | 84s |
| DDA | - | - | Diffusion model | 13,277s |
| Tent | 50,000 | 50,000 | N/A | 110s |
| EATA | 50,000 | 19,608 | regularizer | 118s |
| SAR | 66,418 | 30,488 | Model updates | 164s |
| DeYO | 82,843 | 24,714 | Data augmentation | 144s |
| ReCAP | 50,000 | 19,512 | Eq. 9 | 116s |

# Experiments

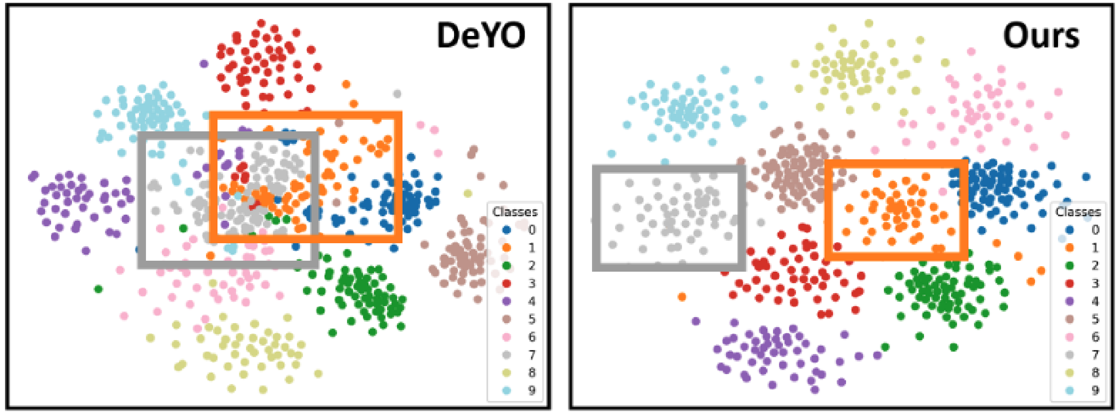



*Figure 5.* The t-SNE visualization of feature space in Snow corruption type after model adaptation.

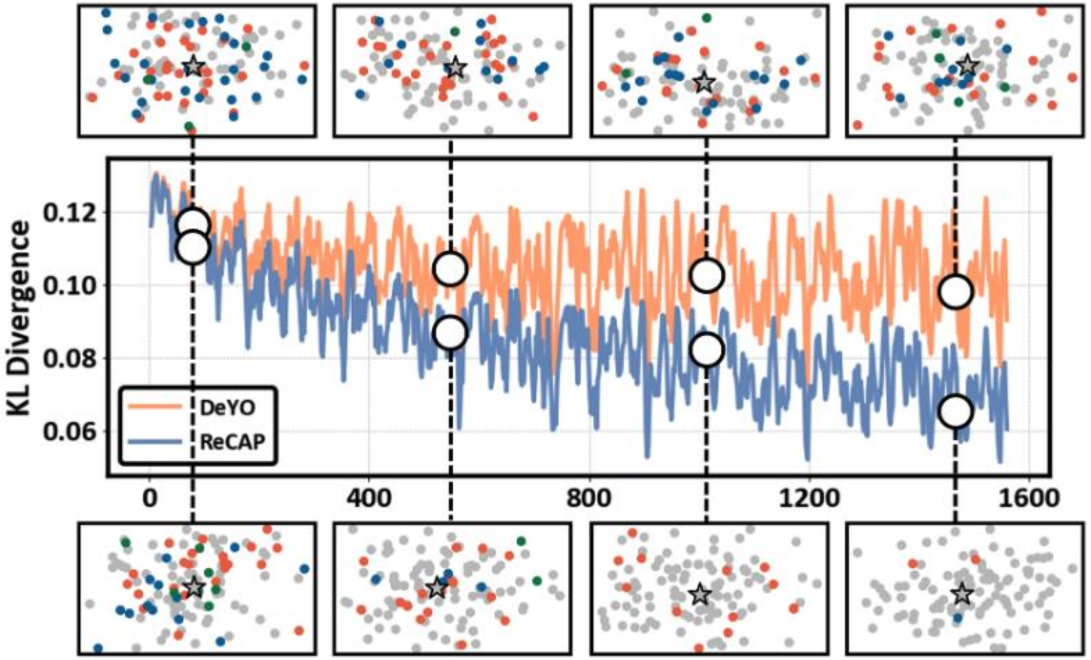



*Figure 6.* Visualization of prediction results and KL Divergence of prediction probabilities in local region. Different colors of points represent different predicted classes of samples within the region.

HAI
Human-centered AI

# Experiments

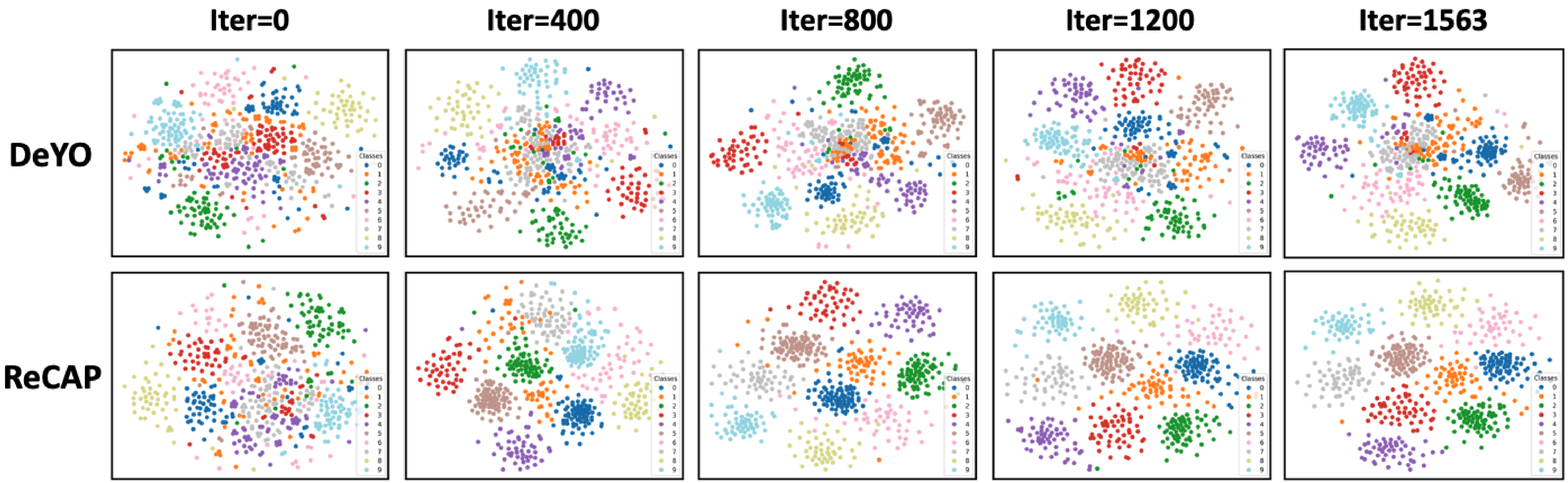



*Figure 8.* The evolution of feature space under DeYO and ReCAP methods. The visualizations are conducted on ImageNet-C under labl shift scenario with ResNet50.


Thank You!